Class 12 Mathematics

Chapter 5: Continuity and Differentiability

Practice Set A: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE 2024 (65/4/1) 1 mark

For what value of k is the function $f(x) = \begin{cases} \dfrac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ continuous at $x = 0$?

(A) 0

(B) $\dfrac{1}{4}$

(C) 1

(D) 4

Formula / idea: Continuous at 0 means $\lim\limits_{x\to 0} f(x) = f(0)$; a $\dfrac{0}{0}$ root form is rationalised.

1 Rationalise the numerator.

$$\frac{\sqrt{4+x}-2}{x}\cdot\frac{\sqrt{4+x}+2}{\sqrt{4+x}+2} = \frac{(4+x)-4}{x\left(\sqrt{4+x}+2\right)} = \frac{1}{\sqrt{4+x}+2}$$

2 Take the limit as $x \to 0$ and equate to $f(0) = k$. ✓ 1

$$\lim_{x\to 0}\frac{1}{\sqrt{4+x}+2} = \frac{1}{2+2} = \frac{1}{4} = k$$

3 So the correct option is (B).

Answer **(B)** $\dfrac{1}{4}$

Watch out: Putting $x = 0$ straight in gives $\dfrac{0}{0}$; rationalise first, do not guess $k = 0$.

2 MCQ CBSE 2024 (65/1/1) 1 mark

Which of the following statements is true for the function $f(x) = \begin{cases} x^2+3, & x \neq 0 \\ 1, & x = 0 \end{cases}$?

(A) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R}$

(B) $f(x)$ is continuous $\forall\, x \in \mathbb{R}$

(C) $f(x)$ **is continuous and differentiable** $\forall\, x \in \mathbb{R} - \{0\}$

(D) $f(x)$ is discontinuous at infinitely many points

Formula / idea: A polynomial is continuous and differentiable; a function is not differentiable where it is discontinuous.

1 Check $x = 0$: compare the limit with $f(0)$.
$\lim_{x \to 0}(x^2+3) = 3 \neq 1 = f(0)$

2 So f is discontinuous at $x = 0$, hence not differentiable there.

3 For $x \neq 0$, $f(x) = x^2 + 3$ is a polynomial: continuous and differentiable. ✓ 1

4 So the correct option is (C).

Answer **(C)** $f(x)$ **is continuous and differentiable** $\forall\, x \in \mathbb{R} - \{0\}$

Watch out: Choosing (B): the single point $x = 0$ breaks continuity because $f(0) = 1$, not 3.

3 MCQ CBSE 2024 (65/4/1) 1 mark

Derivative of e^{2x} with respect to e^x is:

(A) e^x **(B)** $2e^x$

(C) $2e^{2x}$ **(D)** $2e^{3x}$

Formula / idea: Derivative of u w.r.t. v is $\dfrac{du/dx}{dv/dx}$.

1 Let $u = e^{2x}$ and $v = e^x$. Differentiate both w.r.t. x.
$\dfrac{du}{dx} = 2e^{2x}, \qquad \dfrac{dv}{dx} = e^x$

2 Divide. ✓ 1
$\dfrac{du}{dv} = \dfrac{2e^{2x}}{e^x} = 2e^x$

3 So the correct option is (B).

Answer **(B)** $2e^x$

Watch out: Giving $2e^{2x}$ (option C), which is the derivative w.r.t. x, not w.r.t. e^x.

4 MCQ CBSE 2024 (65/1/1) 1 mark

The derivative of $\sin(x^2)$ w.r.t. x, at $x = \sqrt{\pi}$ is:

(A) 1 **(B)** -1

(C) $-2\sqrt{\pi}$ **(D)** $2\sqrt{\pi}$

Formula / idea: Chain rule: $\frac{d}{dx}\sin u = \cos u \cdot \frac{du}{dx}$.

1 Differentiate using the chain rule.
$\frac{d}{dx}\sin(x^2) = \cos(x^2) \cdot 2x$

2 Put $x = \sqrt{\pi}$, so $x^2 = \pi$. ✓ 1
$2\sqrt{\pi}\cos\pi = 2\sqrt{\pi}\,(-1) = -2\sqrt{\pi}$

3 So the correct option is (C).

Answer **(C)** $-2\sqrt{\pi}$

Watch out: Forgetting $\cos\pi = -1$ gives $2\sqrt{\pi}$ (option D).

5 MCQ CBSE 2024 (65/3/1) 1 mark

The derivative of 2^x w.r.t. 3^x is:

(A) $\left(\frac{3}{2}\right)^x \frac{\log 2}{\log 3}$

(B) $\left(\frac{2}{3}\right)^x \frac{\log 3}{\log 2}$

(C) $\left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$

(D) $\left(\frac{3}{2}\right)^x \frac{\log 3}{\log 2}$

Formula / idea: Treat x as the parameter: $\frac{du}{dv} = \frac{du/dx}{dv/dx}$; $\frac{d}{dx}a^x = a^x \log a$.

1 Let $u = 2^x$, $v = 3^x$. Differentiate both w.r.t. x.
$\frac{du}{dx} = 2^x \log 2, \qquad \frac{dv}{dx} = 3^x \log 3$

2 Divide. ✓ 1
$\frac{du}{dv} = \frac{2^x \log 2}{3^x \log 3} = \left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$

3 So the correct option is (C).

Answer **(C)** $\left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$

Watch out: Dividing the wrong way round gives option (D).

6 **Assertion-Reason** CBSE SQP 2024-25 1 mark

Assertion (A): Consider the function defined as $f(x) = |x| + |x-1|$, $x \in \mathbb{R}$. Then $f(x)$ is not differentiable at $x = 0$ and $x = 1$. **Reason (R):** Suppose f be defined and continuous on (a, b) and $c \in (a, b)$; then $f(x)$ is not differentiable at $x = c$ if $\lim\limits_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \neq \lim\limits_{h \to 0^+} \frac{f(c+h) - f(c)}{h}$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Differentiable at c needs LHD $=$ RHD.

1 Remove the modulus near 0 and 1.

$$f(x) = \begin{cases} 1 - 2x, & x < 0 \\ 1, & 0 \leq x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$$

2 At $x = 0$: LHD $= -2$, RHD $= 0$; unequal.

Slopes of $1 - 2x$ and 1

3 At $x = 1$: LHD $= 0$, RHD $= 2$; unequal. So **A is true**.

Slopes of 1 and $2x - 1$

4 R is the definition: unequal one-sided derivatives mean not differentiable. **R is true**.

5 A follows exactly from the test in R, so **R explains A**. ✓ 1

6 So the correct option is (A).

Answer **(A) Both A and R are true and R is the correct explanation of A**

Watch out: Thinking a continuous function must be differentiable; f is continuous but has corners at 0 and 1.

7 **Short answer** CBSE 2024 (65/2/1) — 2 marks

If $y = \operatorname{cosec}(\cot^{-1} x)$, then prove that $\sqrt{1+x^2}\,\dfrac{dy}{dx} - x = 0$.

Formula / idea: Put $\cot^{-1} x = \theta$ and write $\operatorname{cosec}\theta$ in terms of $\cot\theta$.

1	Let $\cot^{-1} x = \theta$, so $\cot\theta = x$ and $\theta \in (0, \pi)$. $\operatorname{cosec}\theta = \sqrt{1+\cot^2\theta} = \sqrt{1+x^2}$ $\sin\theta > 0$ on $(0,\pi)$, so take the positive root	✓ 1
2	So $y = \sqrt{1+x^2}$. Differentiate w.r.t. x. $\dfrac{dy}{dx} = \dfrac{2x}{2\sqrt{1+x^2}} = \dfrac{x}{\sqrt{1+x^2}}$	✓ ½
3	Multiply by $\sqrt{1+x^2}$. Hence proved. $\sqrt{1+x^2}\,\dfrac{dy}{dx} = x \;\Rightarrow\; \sqrt{1+x^2}\,\dfrac{dy}{dx} - x = 0$	✓ ½

Answer $\sqrt{1+x^2}\,\dfrac{dy}{dx} - x = 0$ **(proved)**

Watch out: Differentiating $\operatorname{cosec}(\cot^{-1} x)$ by the chain rule directly is long; simplify y first.

Level 2

18 marks

8 **Short answer** PYQ-style — 2 marks

Find the value of k for which the function $f(x) = \begin{cases} \dfrac{k\cos x}{\pi - 2x}, & x \neq \dfrac{\pi}{2} \\ 5, & x = \dfrac{\pi}{2} \end{cases}$ is continuous at $x = \dfrac{\pi}{2}$.

Formula / idea: Continuity at $\dfrac{\pi}{2}$: $\lim\limits_{x\to\pi/2} f(x) = f\left(\dfrac{\pi}{2}\right)$; use $\lim\limits_{h\to 0}\dfrac{\sin h}{h} = 1$.

1	Put $x = \dfrac{\pi}{2} - h$; as $x \to \dfrac{\pi}{2}$, $h \to 0$. $\cos\left(\dfrac{\pi}{2} - h\right) = \sin h, \qquad \pi - 2x = 2h$	✓ ½
2	Find the limit. $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{k\cos x}{\pi - 2x} = \lim\limits_{h\to 0}\dfrac{k\sin h}{2h} = \dfrac{k}{2}$	✓ 1

3 Equate to $f\left(\frac{\pi}{2}\right) = 5$. ✓ ½

$$\frac{k}{2} = 5 \;\Rightarrow\; k = 10$$

Answer $k = 10$

Watch out: Equating k (not $\frac{k}{2}$) to 5; the factor 2 from $\pi - 2x = 2h$ is often lost.

9 **Short answer** CBSE SQP 2026-27 3 marks

If $y = x^{\log x} + (\log x)^x$, $x > 1$, then evaluate $\frac{dy}{dx}$.

Formula / idea: Split $y = u + v$; take log of each power separately.

1 Let $u = x^{\log x}$ and $v = (\log x)^x$, so $y = u + v$. ✓ ½

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

log of a sum cannot be split

2 Take log of u and differentiate w.r.t. x.

$$\log u = (\log x)^2 \;\Rightarrow\; \frac{1}{u}\frac{du}{dx} = \frac{2\log x}{x}$$

3 Multiply by u. ✓ 1

$$\frac{du}{dx} = x^{\log x} \cdot \frac{2\log x}{x}$$

4 Take log of v and differentiate (product rule).

$$\log v = x\log(\log x) \;\Rightarrow\; \frac{1}{v}\frac{dv}{dx} = \log(\log x) + x \cdot \frac{1}{\log x} \cdot \frac{1}{x}$$

5 Multiply by v. ✓ 1

$$\frac{dv}{dx} = (\log x)^x \left(\log(\log x) + \frac{1}{\log x}\right)$$

6 Add the two results. ✓ ½

$$\frac{dy}{dx} = x^{\log x} \cdot \frac{2\log x}{x} + (\log x)^x \left(\log(\log x) + \frac{1}{\log x}\right)$$

Answer $\frac{dy}{dx} = x^{\log x} \cdot \frac{2\log x}{x} + (\log x)^x \left(\log(\log x) + \frac{1}{\log x}\right)$

Watch out: Writing $\log u = \log x \cdot \log x$ and then differentiating to $\frac{1}{x}$; it is $(\log x)^2$, giving $\frac{2\log x}{x}$.

10 **Short answer** CBSE 2023 (65/1/3) — 2 marks

If $x = \sqrt{a^{\tan^{-1} t}}$, $y = \sqrt{a^{\cot^{-1} t}}$, then show that $x\dfrac{dy}{dx} + y = 0$.

Formula / idea: Parametric form: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$; take $\log$ first to handle the power.

1 Take log of x and differentiate w.r.t. t. ✓ ½

$$\log x = \frac{1}{2}\tan^{-1} t \, \log a \;\Rightarrow\; \frac{dx}{dt} = \frac{x \log a}{2(1+t^2)}$$

2 Take log of y and differentiate w.r.t. t. ✓ ½

$$\log y = \frac{1}{2}\cot^{-1} t \, \log a \;\Rightarrow\; \frac{dy}{dt} = \frac{-y \log a}{2(1+t^2)}$$

$$\frac{d}{dt}\cot^{-1} t = \frac{-1}{1+t^2}$$

3 Divide. ✓ ½

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{y}{x}$$

4 Cross-multiply. Hence proved. ✓ ½

$$x\frac{dy}{dx} + y = 0$$

Answer $x\dfrac{dy}{dx} + y = 0$ **(proved)**

Watch out: Missing the minus sign in the derivative of $\cot^{-1} t$ gives $x\dfrac{dy}{dx} = y$.

11 **Short answer** CBSE 2024 (65(B)) — 3 marks

If $x = a\cos\theta$ and $y = b\sin\theta$, then prove that $\dfrac{d^2y}{dx^2} = -\dfrac{b^4}{a^2y^3}$.

Formula / idea: $\dfrac{d^2y}{dx^2} = \dfrac{d}{d\theta}\left(\dfrac{dy}{dx}\right)\cdot\dfrac{d\theta}{dx}$.

1 Differentiate x and y w.r.t. θ. ✓ ½

$$\frac{dx}{d\theta} = -a\sin\theta, \qquad \frac{dy}{d\theta} = b\cos\theta$$

2 Divide to get $\dfrac{dy}{dx}$. ✓ ½

$$\frac{dy}{dx} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b}{a}\cot\theta$$

3 Differentiate again w.r.t. x (chain rule through θ). ✓ 1

$$\frac{d^2y}{dx^2} = \frac{b}{a}\operatorname{cosec}^2\theta \cdot \frac{d\theta}{dx} = \frac{b}{a}\operatorname{cosec}^2\theta \cdot \frac{1}{-a\sin\theta}$$

4 Simplify. ✓ ½

$$\frac{d^2y}{dx^2} = -\frac{b}{a^2\sin^3\theta}$$

5 Put $\sin\theta = \frac{y}{b}$. Hence proved. ✓ ½

$$\frac{d^2y}{dx^2} = -\frac{b}{a^2}\cdot\frac{b^3}{y^3} = -\frac{b^4}{a^2y^3}$$

Answer $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2y^3}$ **(proved)**

Watch out: Writing $\frac{d^2y}{dx^2} = \frac{d^2y/d\theta^2}{d^2x/d\theta^2}$; you must multiply by $\frac{d\theta}{dx}$ instead.

12 **Short answer** CBSE 2024 (65/3/1) 3 marks

If $x\cos(p+y) + \cos p\sin(p+y) = 0$, prove that $\cos p\,\frac{dy}{dx} = -\cos^2(p+y)$, where p is a constant.

Formula / idea: Differentiate implicitly, then remove x using the given equation.

1 Differentiate both sides w.r.t. x (product rule on the first term). ✓ 1

$$\cos(p+y) - x\sin(p+y)\frac{dy}{dx} + \cos p\cos(p+y)\frac{dy}{dx} = 0$$

2 Collect the $\frac{dy}{dx}$ terms. ✓ ½

$$\frac{dy}{dx}\left[\cos p\cos(p+y) - x\sin(p+y)\right] = -\cos(p+y)$$

3 From the given equation, find x. ✓ ½

$$x = -\frac{\cos p\sin(p+y)}{\cos(p+y)}$$

4 Put this x in the bracket and simplify. ✓ ½

$$\cos p\cos(p+y) + \frac{\cos p\sin^2(p+y)}{\cos(p+y)} = \frac{\cos p}{\cos(p+y)}$$

$\cos^2 + \sin^2 = 1$ in the numerator

5 So $\frac{dy}{dx}\cdot\frac{\cos p}{\cos(p+y)} = -\cos(p+y)$. Cross-multiply. Hence proved. ✓ ½

$$\cos p\,\frac{dy}{dx} = -\cos^2(p+y)$$

Answer $\cos p\,\frac{dy}{dx} = -\cos^2(p+y)$ **(proved)**

Watch out: Treating $\cos p$ as a variable; p is a constant, so $\cos p$ has derivative 0.

13 **Short answer** CBSE SQP 2026-27 — 2 marks

If $y = \sin(x + y)$ and $\dfrac{dy}{dx} = 0$, then prove that $y = \pm 1$.

Formula / idea: Implicit differentiation: differentiate each side w.r.t. x, treating y as a function of x.

1 Differentiate both sides w.r.t. x (chain rule). ✓ 1

$$\frac{dy}{dx} = \cos(x + y)\left(1 + \frac{dy}{dx}\right)$$

2 Solve for $\dfrac{dy}{dx}$. ✓ ½

$$\frac{dy}{dx} = \frac{\cos(x + y)}{1 - \cos(x + y)}$$

3 Put $\dfrac{dy}{dx} = 0$: then $\cos(x + y) = 0$, so $\sin(x + y) = \pm 1$. ✓ ½

$$y = \sin(x + y) = \pm 1$$

$$\sin^2 + \cos^2 = 1$$

Answer $y = \pm 1$ **(proved)**

Watch out: Writing $\dfrac{d}{dx}\sin(x + y) = \cos(x + y)$ and missing the factor $\left(1 + \dfrac{dy}{dx}\right)$.

14 **Short answer** CBSE 2024 (65/3/1) — 3 marks

Find the value of a and b so that the function f defined as $f(x) = \begin{cases} \dfrac{x-2}{|x-2|} + a, & \text{if } x < 2 \\ a + b, & \text{if } x = 2 \\ \dfrac{x-2}{|x-2|} + b, & \text{if } x > 2 \end{cases}$ is a continuous function.

Formula / idea: For $x < 2$, $|x - 2| = -(x - 2)$; for $x > 2$, $|x - 2| = x - 2$. Then LHL = RHL = $f(2)$.

1 Find the LHL at $x = 2$. ✓ ½

$$\lim_{x \to 2^-}\left(\frac{x-2}{-(x-2)} + a\right) = -1 + a$$

2 Find the RHL at $x = 2$. ✓ ½

$$\lim_{x \to 2^+}\left(\frac{x-2}{x-2} + b\right) = 1 + b$$

3 LHL $= f(2)$. ✓ 1

$$a - 1 = a + b \Rightarrow b = -1$$

4 RHL $= f(2)$.
$1 + b = a + b \Rightarrow a = 1$ ✓ 1

Answer $a = 1, b = -1$

Watch out: Taking $\frac{x-2}{|x-2|} = 1$ on both sides; it is -1 for $x < 2$.

Level 3

25 marks

15 **Short answer** PYQ-style 3 marks

If $y = \left(x + \sqrt{1+x^2}\right)^n$, show that $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - n^2y = 0$.

Formula / idea: Write y_1 in terms of y, clear the root, then differentiate again.

1 Differentiate w.r.t. x (chain rule).
$y_1 = n\left(x + \sqrt{1+x^2}\right)^{n-1}\left(1 + \frac{x}{\sqrt{1+x^2}}\right)$

2 The bracket equals $\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}}$, so:
$y_1 = \frac{n\left(x+\sqrt{1+x^2}\right)^n}{\sqrt{1+x^2}} = \frac{ny}{\sqrt{1+x^2}}$ ✓ 1

3 Cross-multiply.
$\sqrt{1+x^2}\, y_1 = ny$ ✓ ½

4 Differentiate again w.r.t. x (product rule).
$\sqrt{1+x^2}\, y_2 + \frac{x}{\sqrt{1+x^2}} y_1 = ny_1$ ✓ 1

5 Multiply by $\sqrt{1+x^2}$ and use $\sqrt{1+x^2}\, y_1 = ny$.
$(1+x^2)y_2 + xy_1 = n\sqrt{1+x^2}\, y_1 = n^2y$

6 Rearrange. Hence proved.
$(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - n^2y = 0$ ✓ ½

Answer $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - n^2y = 0$ **(proved)**

Watch out: Not spotting $1 + \frac{x}{\sqrt{1+x^2}} = \frac{x+\sqrt{1+x^2}}{\sqrt{1+x^2}}$, which turns y_1 into $\frac{ny}{\sqrt{1+x^2}}$.

16 **Long answer** PYQ-style — 5 marks

If $(a+bx)e^{y/x} = x$, prove that $x^3\dfrac{d^2y}{dx^2} = \left(x\dfrac{dy}{dx} - y\right)^2$.

Formula / idea: Take $\log$ to free y; find $x\dfrac{dy}{dx} - y$ first, then differentiate it.

1	Take log on both sides. $\log(a+bx) + \dfrac{y}{x} = \log x$	✓ ½
2	Make y the subject. $y = x\log x - x\log(a+bx)$	✓ ½
3	Differentiate w.r.t. x (product rule on each term). $\dfrac{dy}{dx} = \log x + 1 - \log(a+bx) - \dfrac{bx}{a+bx}$	✓ 1
4	Multiply by x and subtract y; the log terms cancel. $x\dfrac{dy}{dx} - y = x - \dfrac{bx^2}{a+bx} = \dfrac{ax}{a+bx} \quad \cdots(1)$	✓ 1
5	Differentiate (1) w.r.t. x; the $\dfrac{dy}{dx}$ terms cancel on the left. $x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} - \dfrac{dy}{dx} = \dfrac{a(a+bx) - ax\cdot b}{(a+bx)^2}$	✓ 1
6	Simplify. $x\dfrac{d^2y}{dx^2} = \dfrac{a^2}{(a+bx)^2}$	✓ ½
7	Multiply by x^2 and use (1). Hence proved. $x^3\dfrac{d^2y}{dx^2} = \left(\dfrac{ax}{a+bx}\right)^2 = \left(x\dfrac{dy}{dx} - y\right)^2$	✓ ½

Answer $x^3\dfrac{d^2y}{dx^2} = \left(x\dfrac{dy}{dx} - y\right)^2$ **(proved)**

Watch out: Differentiating $(a+bx)e^{y/x} = x$ directly; the quotient $\dfrac{y}{x}$ makes it messy. Take log first.

17 **Short answer** CBSE 2024 (65/2/2) — 3 marks

If $x^{30}y^{20} = (x+y)^{50}$, prove that $\dfrac{dy}{dx} = \dfrac{y}{x}$.

Formula / idea: Take $\log$ on both sides: powers come down as multipliers.

1	Take log on both sides. $30\log x + 20\log y = 50\log(x+y)$	✓ ½

2 Differentiate both sides w.r.t. x. ✓ 1

$$\frac{30}{x}+\frac{20}{y}\frac{dy}{dx}=\frac{50}{x+y}\left(1+\frac{dy}{dx}\right)$$

3 Collect the $\frac{dy}{dx}$ terms.

$$\left(\frac{20}{y}-\frac{50}{x+y}\right)\frac{dy}{dx}=\frac{50}{x+y}-\frac{30}{x}$$

4 Simplify both sides. ✓ 1

$$\frac{20x-30y}{y(x+y)}\cdot\frac{dy}{dx}=\frac{20x-30y}{x(x+y)}$$

5 Cancel $\frac{20x-30y}{x+y}$ from both sides. Hence proved. ✓ ½

$$\frac{dy}{dx}=\frac{y}{x}$$

Valid when $20x-30y\neq 0$

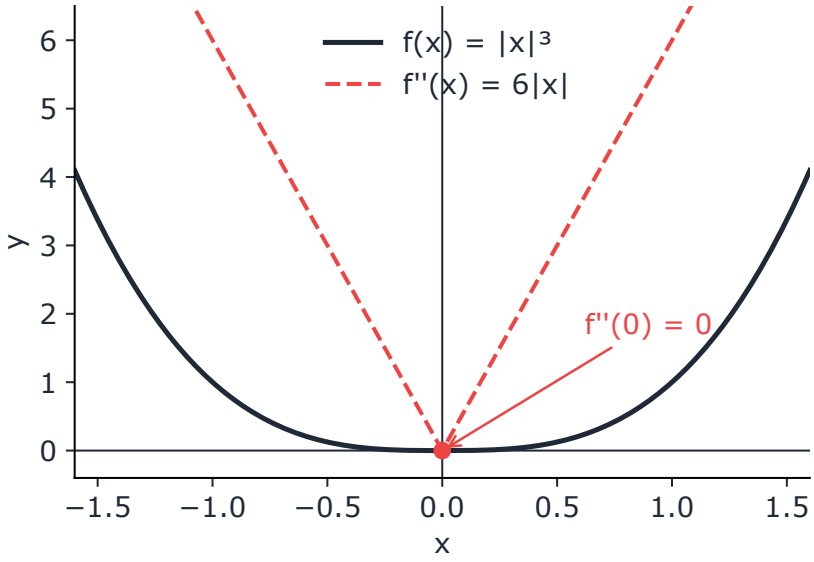
Answer $\frac{dy}{dx}=\frac{y}{x}$ **(proved)**

Watch out: Forgetting the chain-rule factor $\left(1+\frac{dy}{dx}\right)$ when differentiating $\log(x+y)$.

18 **Long answer** CBSE SQP 2024-25 — 5 marks

If $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = |x|^3$, show that $f''(x)$ exists for all real x and find it.

Formula / idea: Remove the modulus; at the joining point $x = 0$ use LHD and RHD, once for f' and once for f''.



1	Remove the modulus. $f(x) = \begin{cases} -x^3, & x < 0 \\ x^3, & x \geq 0 \end{cases}$	✓ 1
2	For $x \neq 0$, differentiate each piece. $f'(x) = -3x^2 \ \ (x < 0), \qquad f'(x) = 3x^2 \ \ (x > 0)$	✓ ½
3	At $x = 0$: find LHD and RHD of f. $\text{LHD} = \lim_{h \to 0} \frac{f(-h) - f(0)}{-h} = \lim_{h \to 0} \frac{h^3}{-h} = 0, \qquad \text{RHD} = \lim_{h \to 0} \frac{h^3}{h} = 0$ $f(-h) = \lvert -h \rvert^3 = h^3$	✓ ½
4	LHD = RHD, so $f'(0) = 0$. $f'(x) = \begin{cases} -3x^2, & x < 0 \\ 3x^2, & x \geq 0 \end{cases}$	✓ 1
5	For $x \neq 0$, differentiate f' again. $f''(x) = -6x \ \ (x < 0), \qquad f''(x) = 6x \ \ (x > 0)$	✓ ½
6	At $x = 0$: find LHD and RHD of f'. $\text{LHD} = \lim_{h \to 0} \frac{f'(-h) - f'(0)}{-h} = \lim_{h \to 0} \frac{-3h^2}{-h} = 0, \qquad \text{RHD} = \lim_{h \to 0} \frac{3h^2}{h} = 0$	
7	LHD = RHD, so $f''(0) = 0$ exists.	✓ 1

8 Hence $f''(x)$ exists for all real x. ✓ ½

$$f''(x) = \begin{cases} -6x, & x < 0 \\ 6x, & x \geq 0 \end{cases} = 6|x|$$

Answer $f''(x) = \begin{cases} -6x, & x < 0 \\ 6x, & x \geq 0 \end{cases}$, **i.e.** $f''(x) = 6|x|$ **for all real** x.

Watch out: Writing $f'(x) = 3x^2$ for all x; for $x < 0$ it is $-3x^2$, since $f(x) = -x^3$ there.

19 **Long answer** CBSE SQP 2026-27 5 marks

If $x = \sin^3 t$, $y = \cos^3 t$, evaluate $\dfrac{d^2y}{dx^2}$ at $\left(\dfrac{1}{2\sqrt{2}}, \dfrac{1}{2\sqrt{2}}\right)$.

Formula / idea: $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \cdot \dfrac{dt}{dx}$.

1 Differentiate x w.r.t. t (chain rule). ✓ ½

$$\frac{dx}{dt} = 3\sin^2 t \cos t$$

2 Differentiate y w.r.t. t. ✓ ½

$$\frac{dy}{dt} = -3\cos^2 t \sin t$$

3 Divide to get $\dfrac{dy}{dx}$. ✓ 1

$$\frac{dy}{dx} = \frac{-3\cos^2 t \sin t}{3\sin^2 t \cos t} = -\cot t$$

4 Differentiate again w.r.t. x (chain rule through t). ✓ 1

$$\frac{d^2y}{dx^2} = \operatorname{cosec}^2 t \cdot \frac{dt}{dx} = \operatorname{cosec}^2 t \cdot \frac{1}{3\sin^2 t \cos t}$$

5 Simplify. ✓ ½

$$\frac{d^2y}{dx^2} = \frac{1}{3\sin^4 t \cos t}$$

6 Find t at the point: $\sin^3 t = \cos^3 t = \dfrac{1}{2\sqrt{2}}$. ✓ ½

$$\sin t = \cos t = \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$$

$$\frac{1}{2\sqrt{2}} = \left(\frac{1}{\sqrt{2}}\right)^3$$

7 Put $\sin^4 t = \frac{1}{4}, \cos t = \frac{1}{\sqrt{2}}$.

$\frac{d^2y}{dx^2} = \frac{1}{3 \cdot \frac{1}{4} \cdot \frac{1}{\sqrt{2}}} = \frac{4\sqrt{2}}{3}$

✓ 1

Answer $\frac{d^2y}{dx^2} = \frac{4\sqrt{2}}{3}$ **at** $\left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)$

Watch out: Differentiating $-\cot t$ to get $\operatorname{cosec}^2 t$ and stopping; you must still multiply by $\frac{dt}{dx}$.

20 **Case study** CBSE 2023 (65/4/1) 4 marks

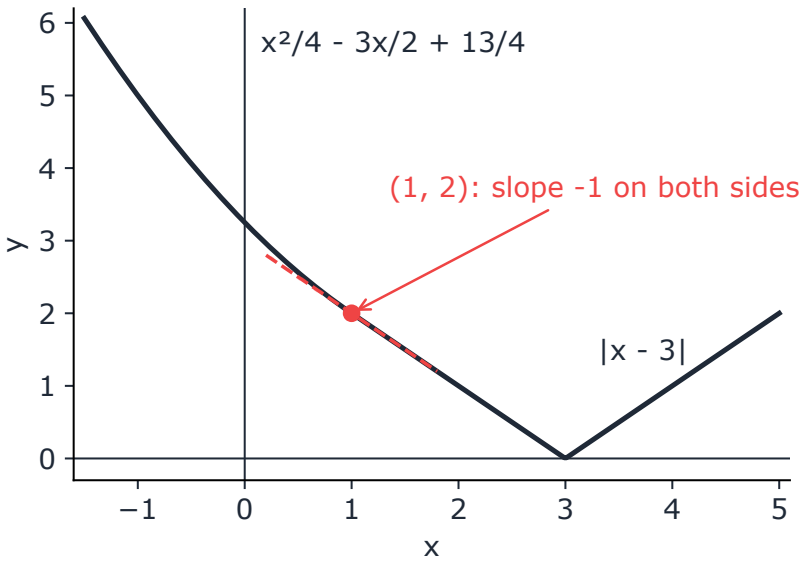
Case Study: Let $f(x)$ be a real valued function. Then its Left Hand Derivative (L.H.D.) is $Lf'(a) = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h}$ and its Right Hand Derivative (R.H.D.) is $Rf'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$. Also, a function $f(x)$ is said to be differentiable at $x = a$ if its L.H.D. and R.H.D. at $x = a$ exist and both are equal. For the function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$ answer the following questions:

(i) What is R.H.D. of $f(x)$ at $x = 1$? 1 mark

(ii) What is L.H.D. of $f(x)$ at $x = 1$? 1 mark

(iii) Check if the function $f(x)$ is differentiable at $x = 1$. 2 marks

Formula / idea: Near $x = 1$ (on the right), $|x-3| = 3 - x$; use the h-definitions given in the case.



(i) 1 mark

1 Find $f(1)$ and $f(1+h)$ from the first piece; $|x-3| = 3 - x$ near 1.

$f(1) = |1-3| = 2, \qquad f(1+h) = 3 - (1+h) = 2 - h$

✓ ½

2 Use the R.H.D. formula. ✓ ½

$$Rf'(1) = \lim_{h\to 0} \frac{(2-h)-2}{h} = \lim_{h\to 0} \frac{-h}{h} = -1$$

Answer (i) **R.H.D. at $x = 1$ is -1.**

(ii)

1 mark

1 Find $f(1-h)$ from the second piece. ✓ ½

$$f(1-h) = \frac{(1-h)^2}{4} - \frac{3(1-h)}{2} + \frac{13}{4} = 2 + h + \frac{h^2}{4}$$

2 Use the L.H.D. formula with $f(1) = 2$. ✓ ½

$$Lf'(1) = \lim_{h\to 0} \frac{h + \frac{h^2}{4}}{-h} = \lim_{h\to 0} \left(-1 - \frac{h}{4}\right) = -1$$

Answer (ii) **L.H.D. at $x = 1$ is -1.**

(iii)

2 marks

1 From (i) and (ii), both one-sided derivatives exist and are equal. ✓ 1

$$Lf'(1) = Rf'(1) = -1$$

2 Hence $f(x)$ is differentiable at $x = 1$. ✓ 1

$$f'(1) = -1$$

Answer (iii) **Yes, f is differentiable at $x = 1$, with $f'(1) = -1$.**

Watch out: Using $|x-3| = x-3$ near $x = 1$; since $x < 3$ there, $|x-3| = 3-x$.